

CSDS 2026 – Week 9

Remark: Using AI to improve the table format, the grammar of answers and multicollinearity explanation for Q3(c).

Initialization

```
set.seed(20260424)
library(compstatslib)
```

Question 1

We will use the `interactive_regression()` function from `CompStatsLib` again

- Scenario 1: Consider a very narrowly dispersed set of points that have a negative or positive steep slope
- Scenario 2: Consider a widely dispersed set of points that have a negative or positive steep slope
- Scenario 3: Consider a very narrowly dispersed set of points that have a negative or positive shallow slope
- Scenario 4: Consider a widely dispersed set of points that have a negative or positive shallow slope

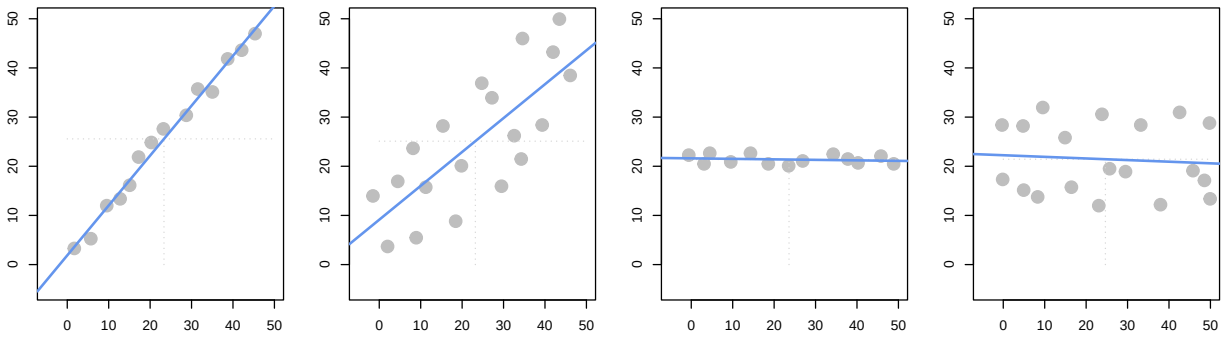
```
# pts1 <- interactive_regression()
# pts2 <- interactive_regression()
# pts3 <- interactive_regression()
# pts4 <- interactive_regression()

# all_pts <- list(
#   scenario1 = pts1,
#   scenario2 = pts2,
#   scenario3 = pts3,
#   scenario4 = pts4
# )

# saveRDS(all_pts, "all_pts.rds")

all_pts <- readRDS("all_pts.rds")
scenario1 <- all_pts$scenario1
scenario2 <- all_pts$scenario2
scenario3 <- all_pts$scenario3
scenario4 <- all_pts$scenario4

par(mfrow = c(1, 4), mar = c(3, 3, 2, 1))
plot_regr(scenario1, stats = FALSE)
plot_regr(scenario2, stats = FALSE)
plot_regr(scenario3, stats = FALSE)
plot_regr(scenario4, stats = FALSE)
```



a. Comparing scenarios 1 and 2, which do we expect to have a stronger R^2 ?

I expect Scenario 1 to have a stronger R^2 than Scenario 2 because the points are more closely clustered around the regression line, indicating a stronger linear relationship between x and y.

The results also confirm this expectation:

```
s1_r2 <- summary(lm(y ~ x, data = scenario1))$r.squared
s2_r2 <- summary(lm(y ~ x, data = scenario2))$r.squared
cat("Scenario 1 R2:", s1_r2, "\n")
## Scenario 1 R2: 0.9856972
cat("Scenario 2 R2:", s2_r2)
## Scenario 2 R2: 0.5770011
```

b. Comparing scenarios 3 and 4, which do we expect to have a stronger R^2 ?

I expect Scenario 3 to have a stronger R^2 than Scenario 4 because the points are more closely clustered around the regression line, indicating a stronger linear relationship between x and y.

The results also confirm this expectation:

```
s3_r2 <- summary(lm(y ~ x, data = scenario3))$r.squared
s4_r2 <- summary(lm(y ~ x, data = scenario4))$r.squared
cat("Scenario 3 R2:", s3_r2, "\n")
## Scenario 3 R2: 0.03369184
cat("Scenario 4 R2:", s4_r2)
## Scenario 4 R2: 0.00650934
```

c. Comparing scenarios 1 and 2, which do we expect has bigger/smaller SSE, SSR, and SST? (intuitively)

I expect:

- SSE: **Scenario 1 will have a smaller SSE** than Scenario 2 because the points are more closely clustered around the regression line, resulting in smaller residuals.
- SSR: In this case, we expect that **Scenario 1 has a slightly larger SSR** than Scenario 2 because its fitted line explains somewhat more of the variation in y. However, since both scenarios show a steep positive linear trend, their SSR values should not differ by very much. The fitted lines pass through roughly the similar lower-left and upper-right regions, so the amount of variation explained by the regression line is fairly similar in both cases.

- SST: I expect SST would likely be *slightly larger in Scenario 2* because SST measures the total variation in the observed y values. Although both scenarios have similar fitted trends, **Scenario 2** shows much greater scatter around the regression line, which increases the overall variability of the data.

d. Comparing scenarios 3 and 4, which do we expect has bigger/smaller SSE, SSR, and SST? (intuitively)

I expect:

- SSE: **Scenario 3 will have a smaller SSE** than Scenario 4 because the points are more closely clustered around the regression line, resulting in smaller residuals.
- SSR: Scenarios 3 and 4 would likely have *similar SSR* values because both show a shallow linear trend
- SST: I expect SST would be *larger in Scenario 4* because SST measures the total variation in the observed y values. Although both scenarios have similar fitted trends, **Scenario 4** shows much greater scatter around the regression line, which increases the overall variability of the data.

Question 2

Let's analyze the `programmer_salaries.txt` dataset we saw in class.

Read the file using `read.csv("programmer_salaries.txt", sep="\t")` because the columns are separated by tabs (`\t`).

```
library(data.table)
salaries <- fread("programmer_salaries.txt", sep = "\t")
head(salaries)
```

```
##      Experience Score Degree Salary
##      <int> <int>  <int>  <num>
## 1:         4    78     0   24.0
## 2:         7   100     1   43.0
## 3:         1    86     0   23.7
## 4:         5    82     1   34.3
## 5:         8    86     1   35.8
## 6:        10    84     1   38.0
```

a. Use the `lm()` function to estimate the regression model `Salary ~ Experience + Score + Degree`. Show the beta coefficients, R^2 , and the first 5 values of fitted values and residuals

```
model <- lm(Salary ~ Experience + Score + Degree, data = salaries)
```

```
# Beta Coefficients
```

```
model$coefficients
```

```
## (Intercept) Experience      Score      Degree
##   7.944849   1.147582   0.196937   2.280424
```

```
# R Squared
```

```
summary(model)$r.squared
```

```
## [1] 0.8467961
```

```
# First 5 Fitted Values
```

```
fitted(model)[1:5]
```

```
##      1      2      3      4      5
## 27.89626 37.95204 26.02901 32.11201 36.34251
```

```
# First 5 Residuals
```

```
resid(model)[1:5]
```

```
##      1      2      3      4      5
## -3.8962605  5.0479568 -2.3290112  2.1879860 -0.5425072
```

b. Use only linear algebra and the geometric view of regression to estimate the regression yourself:

- i. Create an X matrix that has a first column of 1s followed by columns of the independent variables (*only show the code*)

```
X <- as.matrix(cbind(
  Intercept = 1,
  Experience = salaries$Experience,
  Score = salaries$Score,
  Degree = salaries$Degree
))
```

- ii. Create a y vector with the Salary values (*only show the code*)

```
y <- as.matrix(salaries$Salary)
```

- iii. Compute the beta_hat vector of estimated regression coefficients (*show the code and values*)

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (1)$$

```
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
beta_hat
```

```
##      [,1]
## Intercept  7.944849
## Experience  1.147582
## Score      0.196937
## Degree     2.280424
```

- iv. Using the above, compute a y_hat vector of estimated values, and a res vector of residuals (*show the code and the first 5 values of y_hat and res*)

$$\hat{y} = X\hat{\beta} \quad (2)$$

$$\varepsilon = y - \hat{y} \quad (3)$$

```
y_hat <- X %*% beta_hat
res <- y - y_hat
```

```
# First 5 Estimated Values
```

```
y_hat[1:5]
```

```
## [1] 27.89626 37.95204 26.02901 32.11201 36.34251
```

```
# First 5 Residuals
```

```
res[1:5]
```

```
## [1] -3.8962605  5.0479568 -2.3290112  2.1879860 -0.5425072
```

- v. Using only the results from (i) – (iv), compute SSR, SSE and SST (*show the code and values*)

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \quad (4)$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (5)$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (6)$$

```
y_bar <- mean(y)

SSR <- sum((y_hat - y_bar)^2)
SSE <- sum((y - y_hat)^2)
SST <- sum((y - y_bar)^2)

result_table <- data.frame(
  Measure = c("SSR", "SSE", "SST"),
  Value = c(SSR, SSE, SST)
)

knitr::kable(result_table)
```

Measure	Value
SSR	507.89601
SSE	91.88949
SST	599.78550

c. Compute R^2 for in two ways, and confirm you get the same results (*show code and values*):

- i. Use any combination of SSR, SSE, and SST

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \quad (7)$$

```
r_squared_1 <- SSR / SST
r_squared_2 <- 1 - (SSE / SST)
```

- ii. Use the squared correlation of vectors y and y_{hat}

$$R^2 = \text{cor}(y, \hat{y})^2 \quad (8)$$

```
r_squared_3 <- cor(y, y_hat)^2
```

Below are the results from methods (i) and (ii), which confirm that they are identical:

```
r2_table <- data.frame(
  Method = c(
    "SSR / SST",
    "1 - SSE / SST",
  )
```

```

    "cor(y, y_hat)^2"
  ),
  `R_squared` = c(r_squared_1, r_squared_2, r_squared_3)
)

knitr::kable(r2_table, digits = 6)

```

Method	R_squared
SSR / SST	0.846796
1 - SSE / SST	0.846796
cor(y, y_hat)^2	0.846796

Question 3

We're going to look back at the early days of global car manufacturing, when American, Japanese, and European cars competed to rule the world.

No.	Variable	Description
1	mpg	miles-per-gallon (dependent variable)
2	cylinders	cylinders in engine
3	displacement	size of engine
4	horsepower	power of engine
5	weight	weight of car
6	acceleration	acceleration ability of car
7	model_year	year model was released
8	origin	place car was designed (1: USA, 2: Europe, 3: Japan)
9	car_name	make and model names

```

auto <- read.table("auto-data.txt", header=FALSE, na.strings = "?")
auto <- na.omit(auto)
names(auto) <- c("mpg", "cylinders", "displacement", "horsepower", "weight",
                 "acceleration", "model_year", "origin", "car_name")

```

a. Let's first try exploring this data and problem:

- i. Visualize the data as you wish to answer the following questions (report only relevant plots)

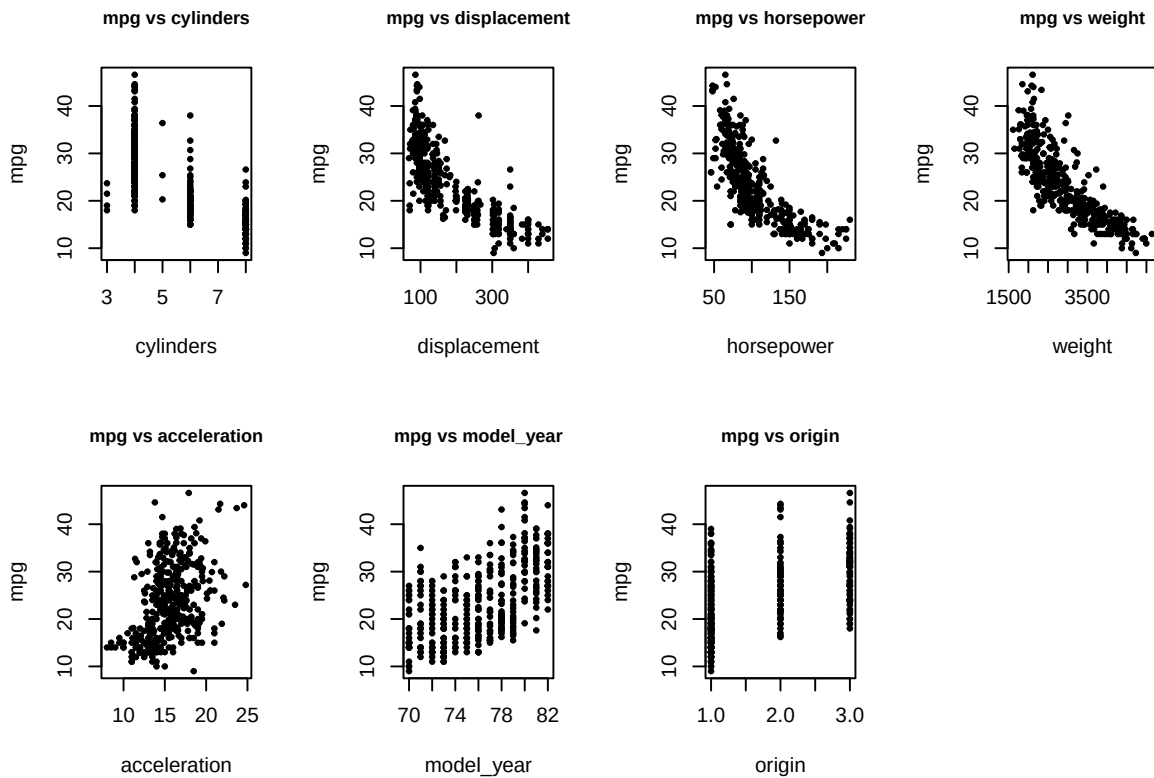
```

par(mfrow = c(2, 4), cex.main = 0.9)

vars <- c("cylinders", "displacement", "horsepower", "weight",
          "acceleration", "model_year", "origin")

for (v in vars) {
  plot(
    auto[[v]], auto$mpg,
    xlab = v, ylab = "mpg",
    main = paste("mpg vs", v),
    pch = 19, cex = 0.5
  )
}

```



- ii. Report a correlation table of all variables, rounding to two decimal places

```
numeric_vars <- auto[, sapply(auto, is.numeric)]
correlation_matrix <- round(cor(numeric_vars), 2)
knitr::kable(correlation_matrix)
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin
mpg	1.00	-0.78	-0.81	-0.78	-0.83	0.42	0.58	0.57
cylinders	-0.78	1.00	0.95	0.84	0.90	-0.50	-0.35	-0.57
displacement	-0.81	0.95	1.00	0.90	0.93	-0.54	-0.37	-0.61
horsepower	-0.78	0.84	0.90	1.00	0.86	-0.69	-0.42	-0.46
weight	-0.83	0.90	0.93	0.86	1.00	-0.42	-0.31	-0.59
acceleration	0.42	-0.50	-0.54	-0.69	-0.42	1.00	0.29	0.21
model_year	0.58	-0.35	-0.37	-0.42	-0.31	0.29	1.00	0.18
origin	0.57	-0.57	-0.61	-0.46	-0.59	0.21	0.18	1.00

- iii. From the visualizations and correlations, which variables appear to relate to mpg?
Among the numeric variables, cylinders, displacement, horsepower, and weight appear to have a negative relationship with mpg, while model_year show a positive relationship. The variable origin also seems to have a positive relationship with mpg.
- iv. Which relationships might not be linear? (*don't worry about linearity for rest of this HW*)
 - The relationships between mpg and cylinders, horsepower, weight, and displacement appear to be potentially non-linear, as the scatterplots suggest some curvature.

- The relationship with `acceleration` appears less clearly linear because the points are more widely scattered and do not follow a strong straight-line pattern.
- Since `origin` is a categorical variable, its relationship with `mpg` is not naturally represented by a simple linear pattern.
- v. **Are there any pairs of *independent variables* that are highly correlated ($r > 0.7$)?**
 Since the question asks for highly correlated pairs and does not restrict this to positive correlations only, I consider the absolute value of the correlation and identify pairs with $|r| > 0.7$ (though the results are all positive).

```
# Exclude the dependent variable 'mpg' from the correlation matrix
# focus on independent variables only
predictor_vars <- numeric_vars[, colnames(numeric_vars) != "mpg"]
predictor_corr_matrix <- round(cor(predictor_vars), 2)

high_corr_idx <- which(
  abs(predictor_corr_matrix) > 0.7 & upper.tri(predictor_corr_matrix),
  arr.ind = TRUE
)

high_corr_pairs <- data.frame(
  Var1 = rownames(predictor_corr_matrix)[high_corr_idx[, 1]],
  Var2 = colnames(predictor_corr_matrix)[high_corr_idx[, 2]],
  Correlation = predictor_corr_matrix[high_corr_idx]
)

high_corr_pairs <- high_corr_pairs[
  order(-abs(high_corr_pairs$Correlation)),
]

knitr::kable(high_corr_pairs)
```

	Var1	Var2	Correlation
1	cylinders	displacement	0.95
5	displacement	weight	0.93
3	displacement	horsepower	0.90
4	cylinders	weight	0.90
6	horsepower	weight	0.86
2	cylinders	horsepower	0.84

b. Let's create a linear regression model where `mpg` is dependent upon all other suitable variables

Note: 'origin' is categorical with three levels, so use 'factor(origin)' in 'lm(...)' to split it into two dummy variables

- i. Which independent variables have a 'significant' relationship with `mpg` at 1% significance?

```
model_mpg <- lm(mpg ~ cylinders + displacement + horsepower + weight +
  acceleration + model_year + factor(origin),
  data = auto)
summary(model_mpg)
```



```
##
## Call:
## lm(formula = mpg ~ cylinders + displacement + horsepower + weight +
##      acceleration + model_year + factor(origin), data = auto)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.0095 -2.0785 -0.0982  1.9856 13.3608
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.795e+01  4.677e+00  -3.839 0.000145 ***
## cylinders    -4.897e-01  3.212e-01  -1.524 0.128215
## displacement  2.398e-02  7.653e-03   3.133 0.001863 **
## horsepower   -1.818e-02  1.371e-02  -1.326 0.185488
## weight       -6.710e-03  6.551e-04 -10.243 < 2e-16 ***
## acceleration  7.910e-02  9.822e-02   0.805 0.421101
## model_year    7.770e-01  5.178e-02 15.005 < 2e-16 ***
## factor(origin)2 2.630e+00  5.664e-01   4.643 4.72e-06 ***
## factor(origin)3 2.853e+00  5.527e-01   5.162 3.93e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.307 on 383 degrees of freedom
## Multiple R-squared:  0.8242,    Adjusted R-squared:  0.8205
## F-statistic: 224.5 on 8 and 383 DF,  p-value: < 2.2e-16
```

Based on the result, the independent variables that have a significant relationship with mpg at the 1% significance level are:

- displacement
- weight
- model_year
- origin (captured by the dummy variables created from factor(origin); factor(origin)2 represents Europe and factor(origin)3 represents Japan, and both are statistically significant relative to the baseline category, USA)

- ii. Looking at the coefficients, is it possible to determine which independent variables are the most effective at increasing mpg? If so, which ones, and if not, why not?

hint: units!

It is not appropriate to determine which variables are most effective at increasing mpg by comparing the raw coefficients alone, because the predictors are measured in different units and origin is a categorical variable. The coefficients can show the direction of the relationship, but not the relative importance of the variables.

c. Let's try to resolve some of the issues with our regression model above.

- i. Create *fully standardized* regression results: are these slopes easier to compare?

Note: consider if you should standardize 'origin'

```
auto_std <- data.frame(scale(auto[1:7]), origin = factor(auto$origin))
model_mpg_std <- lm(mpg ~ ., data = auto_std)
summary(model_mpg_std)
```

```
##
## Call:
```

```
## lm(formula = mpg ~ ., data = auto_std)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.15432 -0.26630 -0.01259  0.25440  1.71182
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.13213     0.03155  -4.187 3.50e-05 ***
## cylinders    -0.10703     0.07020  -1.524  0.12821
## displacement  0.32149     0.10261   3.133  0.00186 **
## horsepower   -0.08967     0.06761  -1.326  0.18549
## weight       -0.73028     0.07130 -10.243 < 2e-16 ***
## acceleration  0.02796     0.03472   0.805  0.42110
## model_year    0.36673     0.02444  15.005 < 2e-16 ***
## origin2       0.33696     0.07257   4.643 4.72e-06 ***
## origin3       0.36556     0.07082   5.162 3.93e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4236 on 383 degrees of freedom
## Multiple R-squared:  0.8242,    Adjusted R-squared:  0.8205
## F-statistic: 224.5 on 8 and 383 DF,  p-value: < 2.2e-16
```

These slopes are easier to compare for the continuous predictors because they come from a standardized model. A larger absolute coefficient means a larger change in mpg in standard deviation units for a one standard deviation increase in that predictor, holding the others fixed. From this table, **weight** appears to have the strongest association in magnitude, followed by **model_year** and **displacement**.

However, **origin2** and **origin3** are dummy variables, so their coefficients are not directly comparable to the standardized continuous predictors in quite the same way.

- ii. Regress mpg over each nonsignificant independent variable, individually. Which ones become significant when we regress mpg over them individually?

```
nonsig_vars <- c("cylinders", "horsepower", "acceleration")

for (var in nonsig_vars) {
  cat("\n===== \n")
  cat("Variable:", var, "\n")
  cat("===== \n")

  model_individual <- lm(as.formula(paste("mpg ~", var)), data = auto)
  print(summary(model_individual))
}

##
## =====
## Variable: cylinders
## =====
##
## Call:
## lm(formula = as.formula(paste("mpg ~", var)), data = auto)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.2413  -3.1832  -0.6332   2.5491  17.9168
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  42.9155     0.8349   51.40  <2e-16 ***
## cylinders    -3.5581     0.1457  -24.43  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.914 on 390 degrees of freedom
## Multiple R-squared:  0.6047,    Adjusted R-squared:  0.6037
## F-statistic: 596.6 on 1 and 390 DF,  p-value: < 2.2e-16
##
##
## =====
## Variable: horsepower
## =====
##
## Call:
## lm(formula = as.formula(paste("mpg ~", var)), data = auto)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5710  -3.2592  -0.3435   2.7630  16.9240
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  39.935861   0.717499   55.66  <2e-16 ***
## horsepower  -0.157845   0.006446  -24.49  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.906 on 390 degrees of freedom
## Multiple R-squared:  0.6059,    Adjusted R-squared:  0.6049
## F-statistic: 599.7 on 1 and 390 DF,  p-value: < 2.2e-16
##
##
## =====
## Variable: acceleration
## =====
##
## Call:
## lm(formula = as.formula(paste("mpg ~", var)), data = auto)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.989  -5.616  -1.199   4.801  23.239
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.8332     2.0485   2.359  0.0188 *
## acceleration   1.1976     0.1298   9.228  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 7.08 on 390 degrees of freedom
## Multiple R-squared:  0.1792,    Adjusted R-squared:  0.1771
## F-statistic: 85.15 on 1 and 390 DF,  p-value: < 2.2e-16
```

When regressing mpg over each of the nonsignificant independent variables (`cylinders`, `horsepower` and `acceleration`) individually, I found that every of them becomes significant at the 1% level. `cylinders` and `horsepower` have a stronger relationship with mpg with about 60% R-squared, while `acceleration` has a weaker relationship with mpg with about 10% R-squared.

This suggests that the lack of significance in the full model for these variables may be due to multicollinearity, since these variables are highly correlated with other independent variables. As a result, their standard errors can be inflated, making it harder to detect significant relationships when other correlated predictors are included in the model.

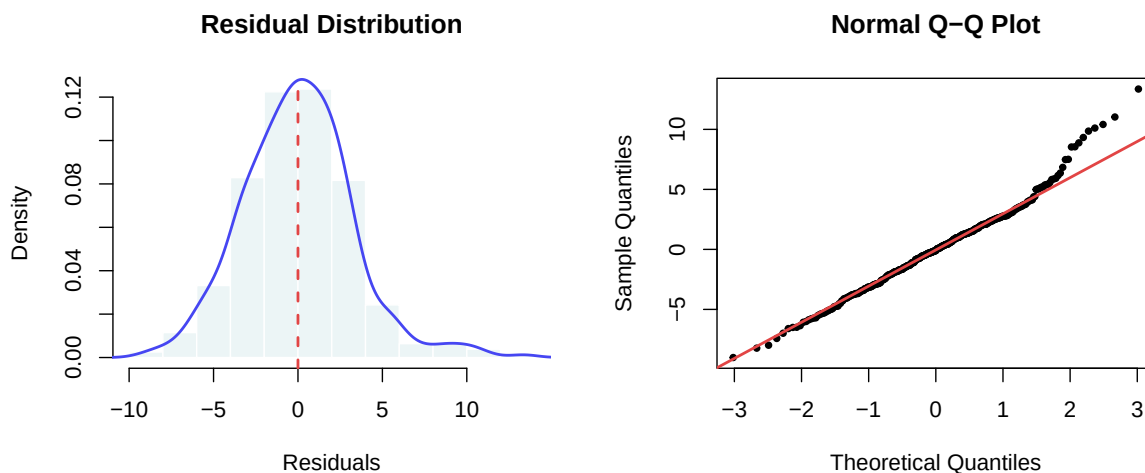
- iii. **Plot the distribution of the residuals: are they normally distributed and centered around zero?**

```
res <- resid(model_mpg)

par(mfrow = c(1, 2))

hist(res,
     probability = TRUE,
     main = "Residual Distribution",
     xlab = "Residuals",
     col = "#ecf5f6",
     border = "white")
lines(density(res), col = "#4646f2", lwd = 2)
abline(v = 0, col = "#e24545", lwd = 2, lty = 2)

qqnorm(res,
      main = "Normal Q-Q Plot",
      pch = 19, cex = 0.6)
qqline(res, col = "#e24545", lwd = 2)
```



The residuals appear to be centered around zero, since the distribution is concentrated near zero and the Q-Q plot is roughly symmetric in the middle. However, they are not exactly normally distributed, since the Q-Q plot shows noticeable deviation in the upper tail, suggesting some right-skewness or heavier upper-tail behavior.